Constrained Optimization, Lagrange Multipliers and KKT Conditions

Machine Learning 2 — Week 6

Imagine a hiker who wants to reach the lowest point of a valley. Without any restrictions, the answer is easy: keep walking downhill until the ground is flat. Now someone builds fences across the valley, and the lowest *reachable* point may sit right against a fence, on a slope that is still tilted downhill. Everything in this chapter is about finding that point reliably, and the Support Vector Machine is exactly such a hiker with fences.

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1. The SVM Story So Far

We are given a dataset of n labelled points

$$\mathrm{D} = \{(x_1, y_1), \ldots, (x_n, y_n)\}, \qquad y_i \in \{-1, +1\}$$

where $y_i = -1$ marks one class and $y_i = +1$ marks the other. The task is a **binary classification** problem: find a **linear boundary** that separates the two classes **and** maximizes the margin between them. Many lines may separate the classes, but only one of them leaves the widest empty "road" between the two groups.

Geometry of the hard-margin SVM

- θ is the **normal vector** of the boundary.
- b is the **offset (bias)** of the boundary from the origin. The actual distance of the boundary from the origin is $|b|/\|\theta\|$.
- The boundary itself is $x^\top\theta + b = 0$.
- The two margin hyperplanes are $x^\top\theta + b = +1$ and $x^\top\theta + b = -1$.
- The points lying exactly on the margin hyperplanes are the **support vectors**.

Why does maximizing the margin equal minimizing $\theta^\top\theta$? The distance of a point x_0 from the boundary is $\dfrac{|x_0^\top\theta + b|}{\|\theta\|}$. A point on $x^\top\theta + b = 1$ is therefore at distance $\gamma = \dfrac{1}{\|\theta\|}$, and the total margin width is

$$2\gamma = \frac{2}{\|\theta\|}$$

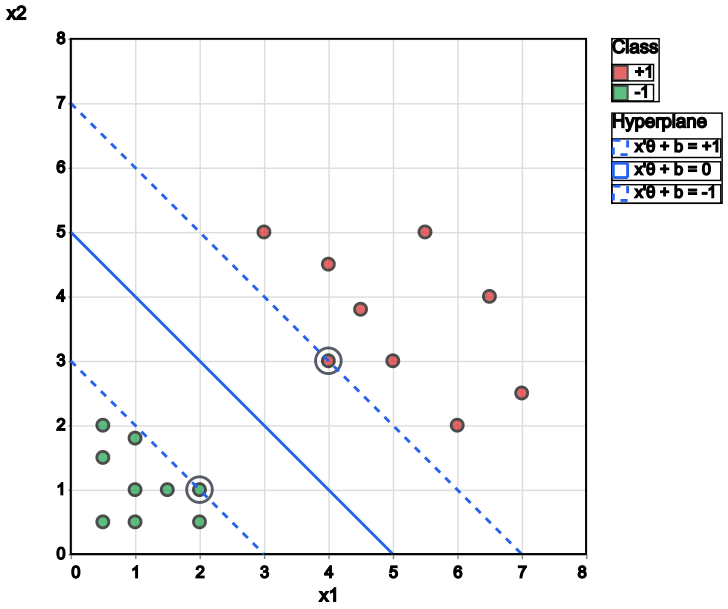
Making 2γ large is the same as making $\|\theta\|$ small, which is the same as making $\theta^\top\theta$ small.

Hard-margin SVM (linearly separable classes)

$$\min_{\theta, b} \ \theta^\top \theta$$

$$\text{s.t.} \quad y_i(\theta^\top x_i + b) \geq 1, \quad i \in \{1, \dots, n\}$$

The constraints make sure that all data points are correctly classified: every positive point ($y_i = +1$) lies beyond the $+1$ line and every negative point ($y_i = -1$) lies beyond the -1 line. Multiplying by y_i merges both statements into one inequality.



Reading the figure: the solid blue line is the decision boundary, the two dashed lines are the margin hyperplanes $x^\top \theta + b = \pm 1$, and the distance between the dashed lines is 2γ. The two circled points (one per class) sit exactly on the dashed lines. They are the **support vectors**, and they alone determine the position of the boundary.

SVM for non-separable classes (soft margin)

Sometimes no line separates the two classes perfectly, because a few red points land on the green side and a few green points land on the red side. If this happens for only a very small fraction of the data, we can still use a linear boundary and treat the offending points as **outliers**.

To allow this, each data point x_i gets a **slack variable** $\eta_i \geq 0$, which is the amount by which that point is allowed to go beyond its margin hyperplane. If a point is on the correct side and outside the margin, $\eta_i = 0$.

But the slacks cannot be free: if we allowed large slack everywhere, any boundary would look "feasible". So the total slack is penalized in the cost, which keeps the amount of violation as small as possible.

$$\min_{\theta, b, \eta} \ \theta^\top \theta + c \sum_{i=1}^{n} \eta_i$$

$$\text{s.t.} \quad y_i(\theta^\top x_i + b) \geq 1 - \eta_i, \quad i \in \{1, \dots, n\}$$

$$\eta_i \geq 0, \quad i \in \{1, \dots, n\}$$

Here c is a **hyperparameter**: it sets the exchange rate between a wide margin (small $\theta^\top \theta$) and few margin violations (small $\sum_i \eta_i$).

	Hard margin	Soft margin
Unknowns	θ, b	θ, b, η
Cost	$\theta^\top \theta$	$\theta^\top \theta + c \sum_i \eta_i$
Constraint	$y_i(\theta^\top x_i + b) \geq 1$	$y_i(\theta^\top x_i + b) \geq 1 - \eta_i,\ \eta_i \geq 0$
Works when	Classes are linearly separable	Always

> **Note for Review** — The soft-margin problem is the *more general* one. If the classes are linearly separable, the optimizer simply returns all $\eta_i = 0$ and you recover the hard-margin SVM. A large c punishes violations heavily (closer to hard margin); a small c tolerates many violations in exchange for a wider margin.

2. Rewriting SVM as a Quadratic Program

Before choosing a solver, we put both SVM problems into one standard shape. The shape is: **quadratic cost, linear inequality constraints**.

2.1 Hard-margin SVM

Define the stacked variable $\omega = (\theta, b)$. The cost $\theta^\top \theta$ has no b in it, so we add a harmless term $0 \cdot b^\top b$ and write the cost as a quadratic form in ω:

$$\theta^\top \theta + 0\, b^\top b = \begin{bmatrix} \theta^\top & b \end{bmatrix} \begin{bmatrix} I & 0 \\ 0^\top & 0 \end{bmatrix} \begin{bmatrix} \theta \\ b \end{bmatrix} = \omega^\top B\, \omega$$

Check by multiplying: first $B\omega = \begin{bmatrix} I\theta + 0 \cdot b \\ 0^\top \theta + 0 \cdot b \end{bmatrix} = \begin{bmatrix} \theta \\ 0 \end{bmatrix}$, then $\omega^\top (B\omega) = \theta^\top \theta + b \cdot 0 = \theta^\top \theta$.

Each constraint becomes a row-vector times ω:

$$y_i(x_i^\top \theta + b) = y_i x_i^\top \theta + y_i b = \begin{bmatrix} y_i x_i^\top & y_i \end{bmatrix} \begin{bmatrix} \theta \\ b \end{bmatrix} \geq 1$$

Stacking all n rows gives the matrix A, and the vector $\mathbf{1}$ is the vector of all ones:

$$A = \begin{bmatrix} y_1 x_1^\top & y_1 \\ y_2 x_2^\top & y_2 \\ \vdots & \vdots \\ y_n x_n^\top & y_n \end{bmatrix}, \qquad A\omega \geq \mathbf{1}$$

So the hard-margin SVM is

$$\min_{\omega}\ \omega^\top B\, \omega \quad \text{s.t.} \quad A\omega \geq \mathbf{1}$$

2.2 Soft-margin SVM

Step 1 — Start from the problem:

$$\min_{\theta, b, \eta}\ \theta^\top \theta + 0\, b^\top b + c \sum_{i=1}^{n} \eta_i \quad \text{s.t.} \quad y_i(x_i^\top \theta + b) \geq 1 - \eta_i \ \ \forall i$$

Step 2 — Turn the sum into an inner product. The slack sum is

$$\sum_{i=1}^{n} \eta_i = \eta_1 + \eta_2 + \cdots + \eta_n = 1 \cdot \eta_1 + 1 \cdot \eta_2 + \cdots + 1 \cdot \eta_n = \begin{bmatrix} 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \vdots \\ \eta_n \end{bmatrix} = \mathbf{1}^\top \eta$$

Step 3 — Write every term, quadratic and linear. To keep dimensions explicit, $\theta \in \mathrm{R}^d$ (d = number of features), $b \in \mathrm{R}$ and $\eta \in \mathrm{R}^n$. Every vector 0 is a zero vector of the right size (the one multiplying θ has size $d \times 1$):

$$\min_{\theta, b, \eta}\ \underbrace{\theta^\top I\, \theta + b^\top 0\, b + \eta^\top 0\, \eta}_{\text{quadratic terms}} + \underbrace{c\, \mathbf{1}^\top \eta + 0^\top \theta + 0 \cdot b}_{\text{linear terms}}$$

Step 4 — Rearrange the constraint. Moving η_i to the left:

$$y_i x_i^\top \theta + y_i b + \eta_i \geq 1$$

Step 5 — Stack everything into one vector. Let $\omega = (\theta, b, \eta) \in \mathrm{R}^{d+1+n}$. The quadratic part is

$$[\theta^\top \quad b^\top \quad \eta^\top] \begin{bmatrix} I & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ b \\ \eta \end{bmatrix} = \omega^\top Q \omega$$

The block sizes are d, 1, n along each side. Multiplication check: $Q\omega = \begin{bmatrix} \theta \\ 0 \\ 0 \end{bmatrix}$, hence $\omega^\top Q \omega = \theta^\top \theta$.

The linear part is

$$r^\top \omega = [0^\top \quad 0 \quad c\,\mathbf{1}^\top] \begin{bmatrix} \theta \\ b \\ \eta \end{bmatrix} = 0^\top \theta + 0 \cdot b + c\,\mathbf{1}^\top \eta = c \sum_{i=1}^{n} \eta_i$$

> **Note for Review** — The order of the blocks in r must match the order in ω. Since $\omega = (\theta, b, \eta)$, the nonzero block $c\,\mathbf{1}^\top$ must sit in the η **position (last)**. Writing it first by accident would penalize θ instead of the slacks.

Step 6 — Stack the constraints. For data point i, the row of A is $[y_i x_i^\top \quad y_i \quad e_i^\top]$, where $e_i \in \mathrm{R}^n$ has a 1 in position i and 0 elsewhere, so that it picks out exactly η_i:

$$[y_i x_i^\top \quad y_i \quad e_i^\top] \begin{bmatrix} \theta \\ b \\ \eta \end{bmatrix} = y_i x_i^\top \theta + y_i b + \eta_i \geq 1$$

All n rows together:

$$A = \begin{bmatrix} y_1 x_1^\top & y_1 & 1 & 0 & \cdots & 0 \\ y_2 x_2^\top & y_2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \ddots & \vdots \\ y_n x_n^\top & y_n & 0 & 0 & \cdots & 1 \end{bmatrix} = [\mathrm{diag}(y) X \quad y \quad I_n]$$

where $X \in \mathrm{R}^{n \times d}$ has the data points $x_i^\top$ as rows.

Step 7 — The final quadratic program:

$$\min_{\omega} \ \omega^\top Q \omega + r^\top \omega \quad \text{s.t.} \quad A\omega \geq \mathbf{1}$$

The cost is **quadratic** and the constraints are **linear inequalities**, so this is a **Quadratic Program (QP) with linear inequality constraints**. Compared with the hard-margin version, the only new ingredient is the linear term $r^\top \omega$, which exists because of the slack variables introduced to handle outliers.

> **Note for Review** — The slack constraints $\eta_i \geq 0$ are also linear inequalities. To be fully complete, append the rows $[0 \quad 0 \quad I_n]\,\omega \geq 0$ below $A\omega \geq \mathbf{1}$. This is why the soft-margin SVM has $2n$ **inequality constraints** in total (n margin constraints and n non-negativity constraints), while the hard-margin SVM has n.

1. Soft-Margin SVM

Starts with the original SVM optimization problem. The model minimizes the squared norm of the weight vector, $\theta^T\theta$, plus a penalty for violations represented by the slack variables η_i.

2. Pad with Zero Blocks

The variables are reorganized into one larger optimization vector ω. The bias b and slack variables η do not contribute to the quadratic part, so their corresponding blocks in the quadratic matrix are zero blocks.

3. Collect the Slack Sum

The summation of slack variables is rewritten as a vector dot product:

$$\sum_i \eta_i = \mathbf{1}^T\eta$$

This allows the SVM objective to be expressed as a standard quadratic term plus a linear term.

4. Stack the Constraints

Each individual SVM constraint:

$$y_i(\theta^T x_i + b) + \eta_i \geq 1$$

is converted into one row of a matrix:

$$[y_i x_i^T \quad y_i \quad e_i^T]\,\omega \geq 1$$

Stacking the constraints for every training example gives:

$$A\omega \geq \mathbf{1}$$

5. Quadratic Program

The final result is the standard QP form:

$$\min_\omega \frac{1}{2}\omega^T Q\omega + r^T\omega$$

$$\text{subject to: } A\omega \geq \mathbf{1}$$

Summary Pipeline:

Original Soft-Margin SVM → Block-Matrix Reformulation → Linear-Term Collection → Constraint Matrix → Standard QP Formulation

The next question is how to *solve* such problems. For that we need the theory of constrained optimization, starting with the simplest case.

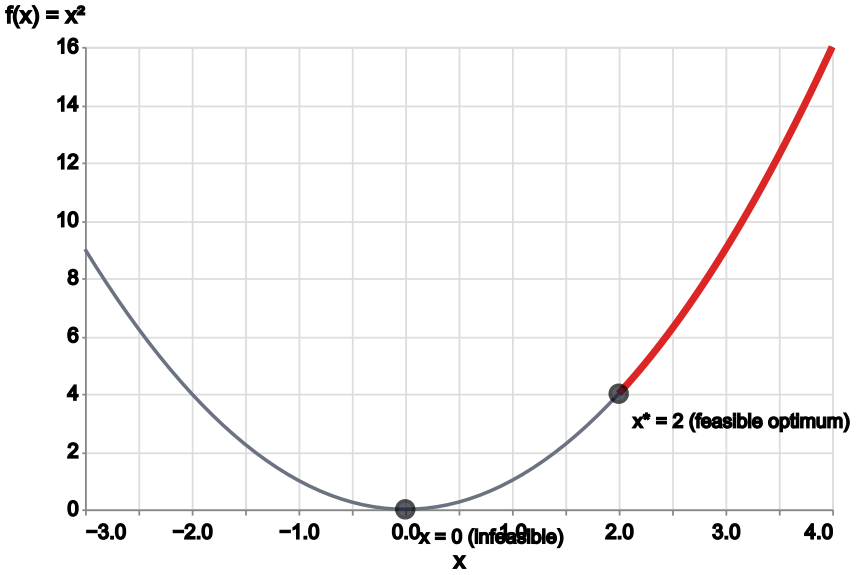
3. Why Setting the Gradient to Zero Fails

Without constraints, minimizing a function is simple: differentiate, set the gradient to zero, solve. Let us see what constraints do to this recipe with the simplest possible function, $f(x) = x^2$.

Unconstrained: $\min_{x \in \mathrm{R}} f(x)$. Here $f'(x) = 2x = 0$ gives $x^* = 0$. Since x may be any real number, this is the answer.

Case 1 — subject to $x = 2$**.** Only one point is feasible, so the optimal solution is trivially $x^* = 2$.

Case 2 — subject to $x \geq 2$**.** Now x may only live in $[2, \infty)$. On this range f keeps increasing as x grows, so the smallest value is at the left edge: $x^* = 2$.



The red segment is the feasible set $x \geq 2$ (the shaded region on the lecture slide). The optimum is at $x^* = 2$, but

$$\nabla f(x)\Big|_{x=2} = 2x\Big|_{x=2} = 4 \neq 0$$

So at a constrained optimum **the gradient of** f **does not have to vanish**. Our old idea "set $\nabla f = 0$" no longer gives the answer. We need a new condition, and the Lagrange multiplier technique supplies it.

4. Equality Constrained Optimization

The general equality-constrained problem is

$$\min_{x} f(x) \quad \text{subject to} \quad g_1(x) = 0,\ g_2(x) = 0,\ \ldots,\ g_k(x) = 0$$

with $f : \mathrm{R}^n \to \mathrm{R}$ and each $g_i : \mathrm{R}^n \to \mathrm{R}$. In words: f and the g_i are multivariable functions that return a single number, and $x \in \mathrm{R}^n$ is an n-dimensional vector that must satisfy **all** k constraints while minimizing f.

> **Note for Review** — The slide indexes the constraints up to m. In these notes the number of equality constraints is k and the number of inequality constraints is m (introduced in Section 8), so that the two letters never clash.

5. The Geometry Behind Lagrange Multipliers

We derive the key condition with a single constraint $g(x) = 0$, step by step.

Step 1 — The feasible set. Let $S = \{x \mid g(x) = 0\}$ be the set of all points that satisfy the constraint (the **feasible set**). For $n = 2$, $g(x) = 0$ describes a curve. For example:

- $g(x) = x_1^2 + x_2^2 - 1 = 0$ is a circle.
- $g(x) = x_1 + x_2 - 1 = 0$ is a line.

Step 2 — Two nearby feasible points. Take $x \in S$ and another point $y \in S$ very close to it, so that $y = x + \Delta x$ with Δx tiny. Locally the curve looks flat, and Δx is a vector pointing along the curve (tangent direction). Because both points are feasible:

$$g(x) = g(y) = 0$$

Step 3 — Taylor expansion of g around x.

$$g(y) = g(x + \Delta x) = g(x) + \nabla g(x)^\top \Delta x + O(\|\Delta x\|_2^2)$$

Since Δx is very small, the higher-order term is negligible. Using $g(y) = g(x) = 0$:

$$\nabla g(x)^\top \Delta x = 0$$

Step 4 — Meaning. The inner product of ∇g and Δx is zero, so ∇g **is perpendicular to the tangent direction** Δx. In other words, at every point, ∇g is perpendicular (normal) to the constraint curve.

Step 5 — What optimality demands of f. Suppose x is an optimal solution and compare $f(x)$ with $f(x + \Delta x)$, where $x + \Delta x$ is another feasible point. To first order,

$$f(x + \Delta x) \approx f(x) + \nabla f(x)^\top \Delta x$$

- If $\nabla f(x)^\top \Delta x < 0$, then $f(x) > f(x + \Delta x)$: we found a feasible point with a smaller value, so x was **not** optimal.
- If $\nabla f(x)^\top \Delta x > 0$, then the opposite direction $-\Delta x$ is also feasible (to first order) and decreases f. Again x was not optimal.

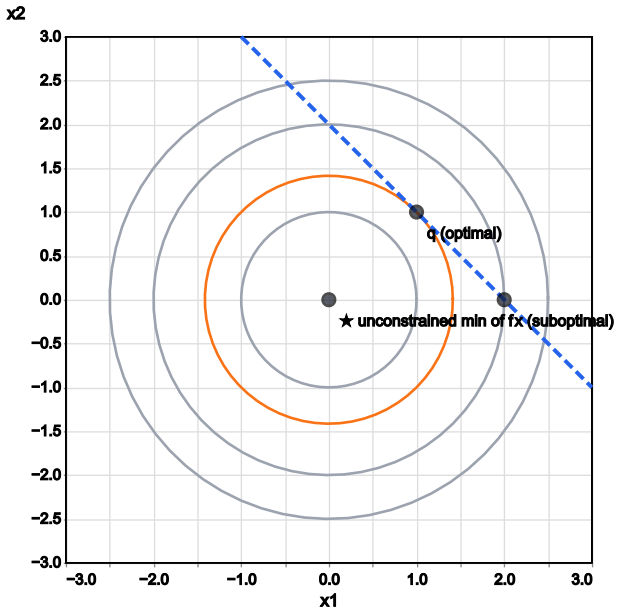
So at an optimum, the rate of change of f along the constraint curve must be zero: $\nabla f(x^*)^\top \Delta x = 0$. Therefore ∇f **is also perpendicular to the tangent direction**. Both ∇f and ∇g are perpendicular to the same tangent line, so they are **parallel** (their magnitudes may differ, hence a scalar λ):

$$\nabla f(x^*) = \lambda \nabla g(x^*)$$

> **Note for Review** — With several constraints, the same argument shows that $\nabla f(x^*)$ must be a linear combination of $\nabla g_1(x^*), \ldots, \nabla g_k(x^*)$, one multiplier per constraint.

Picture: suboptimal vs optimal point

The contours of f are circles centered at the unconstrained minimum $\star$ (here at the origin), and the constraint is the straight line $g(x) = x_1 + x_2 - 2 = 0$. At $x = (2, 0)$ we can slide along the line toward the center and decrease f. The sliding stops exactly where the line **touches** a contour: that point $q = (1, 1)$ is the constrained optimum, and there $\nabla f \parallel \nabla g$.



Legend: gray and orange circles are the contours $f(x) = c$; the dashed blue line is $g(x) = 0$; the orange contour is the one the line just touches.

6. The Lagrangian Recipe

The condition $\nabla f = \lambda \nabla g$ together with $g(x) = 0$ can be produced mechanically. Define the **Lagrange function** (Lagrangian)

$$L(x, \lambda) = f(x) - \lambda\, g(x)$$

The term $\lambda\, g(x)$ is a penalty term attached to the constraint, and λ is the **Lagrange multiplier**. Take the gradient with respect to each argument:

$$\nabla_x L = \nabla f(x) - \lambda\, \nabla g(x) = 0 \quad \Longrightarrow \quad \nabla f = \lambda\, \nabla g$$

$$\nabla_\lambda L = -g(x) = 0 \quad \Longrightarrow \quad g(x^*) = 0$$

The first equation is exactly the parallel-gradients condition we derived geometrically. The second is simply the original constraint. So we **do not** set $\nabla f = 0$ (we saw this fails). We solve the two equations together, for the two unknowns x and λ.

The given optimization process using Lagrange Multipliers can be represented in Markdown as follows:

Optimization via Lagrange Multipliers

1. Problem Formulation

Given:

- Objective: $\min f(x)$
- Subject to: $g(x) = 0$

2. Lagrangian Function

Construct the Lagrangian equation:

$$\mathrm{L}(x, \lambda) = f(x) - \lambda g(x)$$

3. Necessary Conditions (Stationary Points)

Differentiate the Lagrangian to set up the system of equations:

- **Gradient Condition (with respect to x):**

$$\nabla f(x) = \lambda \nabla g(x)$$

- **Constraint Condition (with respect to λ):**

$$g(x) = 0$$

4. Solve System of Equations

Solve the system consisting of $n + k$ equations and $n + k$ unknowns to determine candidate critical points.

5. Candidate Evaluation

Evaluate and compare $f(x)$ at each candidate point to determine whether it corresponds to a minimum, maximum, or saddle point.

7. Worked Example on the Unit Circle

Problem. Minimize $f(x_1, x_2) = x_1 + x_2$ subject to $g(x_1, x_2) = x_1^2 + x_2^2 - 1 = 0$.

Step 1 — Lagrangian.

$$L(x_1, x_2, \lambda) = x_1 + x_2 - \lambda\,(x_1^2 + x_2^2 - 1)$$

Step 2 — Three equations.

$$\frac{\partial L}{\partial x_1} = 1 - 2\lambda x_1 = 0 \tag{1}$$

$$\frac{\partial L}{\partial x_2} = 1 - 2\lambda x_2 = 0 \tag{2}$$

$$\frac{\partial L}{\partial \lambda} = -(x_1^2 + x_2^2 - 1) = 0 \implies x_1^2 + x_2^2 - 1 = 0 \tag{3}$$

Three equations, three unknowns (x_1, x_2, λ).

Step 3 — Solve.

- From (1): $\lambda \neq 0$ (otherwise $1 = 0$), so $x_1 = \dfrac{1}{2\lambda}$.
- From (2): $x_2 = \dfrac{1}{2\lambda}$. Hence $x_1 = x_2$.
- Substitute into (3):

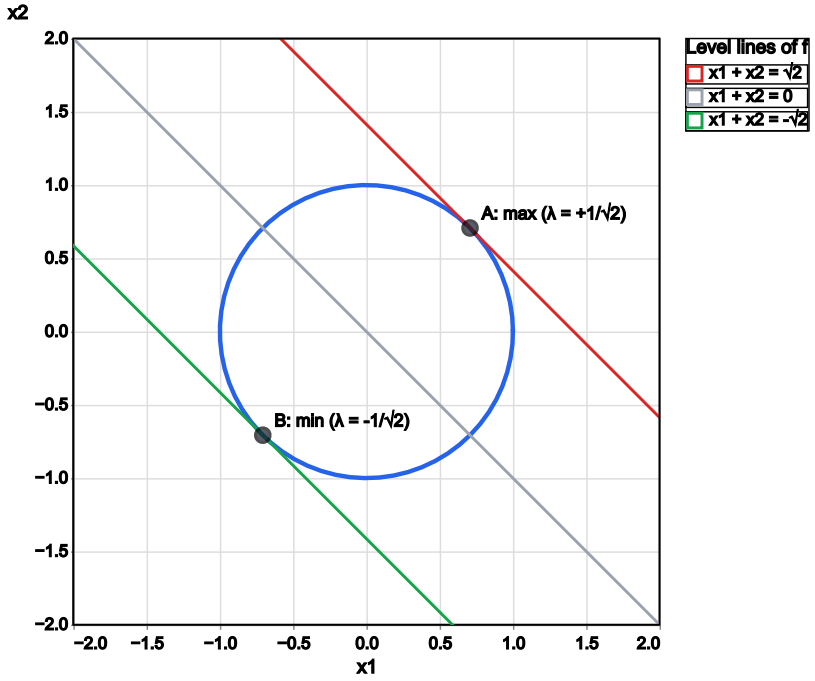
$$\left(\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 = 1 \implies \frac{2}{4\lambda^2} = 1 \implies \lambda^2 = \frac{1}{2} \implies \lambda = \pm\frac{1}{\sqrt{2}}$$

- For $\lambda = +\dfrac{1}{\sqrt{2}}$: $x_1 = x_2 = \dfrac{1}{2/\sqrt{2}} = \dfrac{\sqrt{2}}{2}$.
- For $\lambda = -\dfrac{1}{\sqrt{2}}$: $x_1 = x_2 = -\dfrac{\sqrt{2}}{2}$.

Candidate	λ	$x_1 = x_2$	$f = x_1 + x_2$
A	$+\dfrac{1}{\sqrt{2}}$	$+\dfrac{\sqrt{2}}{2}$	$+\sqrt{2}$
B	$-\dfrac{1}{\sqrt{2}}$	$-\dfrac{\sqrt{2}}{2}$	$-\sqrt{2}$

The solution quoted for this example is $\lambda^* = +\frac{1}{\sqrt{2}}, x_1^* = x_2^* = \frac{\sqrt{2}}{2}$, which is candidate A.

> **Note for Review** — The Lagrange equations only produce **critical points**: they cannot tell a minimum from a maximum. Here both candidates satisfy all three equations. Candidate A ($f = +\sqrt{2}$) is the **largest** value of $x_1 + x_2$ on the circle, while candidate B ($f = -\sqrt{2}$) is the **smallest**. So for the *minimization* problem as stated, the winner after comparing f values is B; A is the answer to the corresponding *maximization*. Pro-Tip: always evaluate f at every candidate and compare.



The level lines of f are parallel straight lines. The red and green ones just touch the circle at the two critical points, which is the tangency condition $\nabla f \parallel \nabla g$ in picture form.

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8. General Constrained Optimization

SVM has **inequality** constraints, so equality constraints alone are not enough. Here is the general setting:

$$f : \mathrm{R}^n \to \mathrm{R}, \qquad g : \mathrm{R}^n \to \mathrm{R}^k, \qquad h : \mathrm{R}^n \to \mathrm{R}^m, \qquad x \in \mathrm{R}^n$$

$$\min_{x \in \mathrm{R}^n} f(x) \quad \text{s.t.} \quad g(x) = 0, \quad h(x) \geq 0$$

- $g(x) = \begin{bmatrix} g_1(x_1, \ldots, x_n) \\ \vdots \\ g_k(x_1, \ldots, x_n) \end{bmatrix}$ stacks the k **equality constraints**, and 0 is a zero vector of size $k \times 1$.
- $h(x) \geq 0$ stacks the m **inequality constraints**, each of the form $h_j(x) \geq 0$, and the 0 here has size $m \times 1$.
- In the SVM, $m = n$ for the hard-margin problem (one inequality per data point).

Equality constraints are handled exactly as before: one penalty term $\lambda_i g_i(x)$ per constraint. With the same idea for inequalities, using multipliers μ_j:

$$L(x, \lambda, \mu) = f(x) - \lambda_1 g_1(x) - \cdots - \lambda_k g_k(x) - \mu_1 h_1(x) - \cdots - \mu_m h_m(x)$$
$$= f(x) - \lambda^\top g(x) - \mu^\top h(x)$$

Its gradient with respect to x is

$$\nabla L = \nabla f - \lambda^\top \nabla g - \mu^\top \nabla h(x) = \nabla f - \sum_{i=1}^{k} \lambda_i \nabla g_i - \sum_{j=1}^{m} \mu_j \nabla h_j = 0$$

(here ∇g and ∇h denote the Jacobian matrices, whose rows are the gradients of the individual constraint functions).

The question: can we simply set this gradient to zero and solve for x? For equality constraints, yes. But with inequalities **not directly**: some inequality constraints will matter for the answer and some will not, and the equation above treats all of them as if they mattered. To fix that, we must understand when an inequality constraint is *active*.

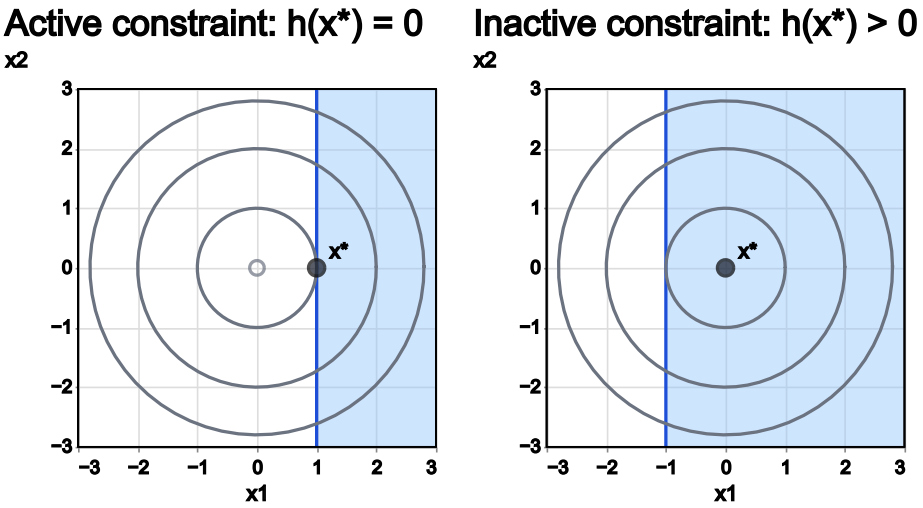
> **Note for Review** — The recipe for solving depends on the type of problem. If f is quadratic and the constraints are linear (a QP, as in the SVM), specialized and efficient algorithms exist. If f is non-convex and the constraints are highly non-linear, the gradient equations are much harder to solve and need different algorithms. These solvers come later; for now the focus is on the conditions an optimal point must satisfy.

9. Active vs Inactive Constraints

Consider one inequality constraint $h(x) \geq 0$.

- $h(x) = 0$ is a curve (the **boundary**).
- $h(x) > 0$ is the whole region on one side of the curve.
- The **feasible set** is the set of all x satisfying the inequality constraints.

There are exactly two situations, depending on where the unconstrained optimum of f lies.



Reading the figure: the shaded half-plane is the feasible set $h(x) > 0$, its edge is $h(x) = 0$, and the gray circles are contours of f. The hollow circle on the left panel is the unconstrained minimum, which lies **outside** the feasible set. (Style after Solomon, 2015.)

Case A — Active constraint: the global optimum is *not* feasible

The unconstrained optimum is cut off by the constraint. The best feasible point must then lie **on the boundary** $h(x) = 0$. The constraint is **active** and the problem behaves like the **equality-constrained** problem $h(x) = 0$. All points strictly inside the region are irrelevant.

Example (from the 1D function): minimize x^2 subject to $x \geq 2$. The global minimizer $x = 0$ is infeasible, so $x^* = 2$, on the boundary.

Case B — Inactive constraint: the global optimum *is* feasible

The unconstrained optimum already satisfies the constraint with room to spare, so the fence changes nothing. The constraint is **inactive** and can be **ignored entirely**: just solve $\nabla f = 0$.

Example: minimize x^2 subject to $x \geq -2$. The global minimizer $x = 0$ satisfies $x \geq -2$, so $x^* = 0$ (and $\nabla f(0) = 0$, as in the unconstrained case).

	Active constraint	Inactive constraint
Is the unconstrained optimum feasible?	No	Yes
Where is x^*?	On the boundary: $h(x^*) = 0$	In the interior: $h(x^*) > 0$
How to treat it	As an equality constraint	Ignore it
Example	$\min x^2$ s.t. $x \geq 2$	$\min x^2$ s.t. $x \geq -2$

The mountain analogy

Picture the hiker from the start of this chapter heading for the lowest point of a valley. If the fences are far away (or not there at all), the hiker reaches the true bottom, and the fences never mattered: they are **inactive**. If there is a fence, say around a volcano, that stops the hiker before the bottom, the hiker ends up standing against the fence: that fence is **active**, and the hiker's position is determined by it. Active fences behave like equality constraints (you are *on* them); inactive ones can be removed without changing the answer.

The inequality constraint decision process can be represented in Markdown as follows:

Inequality Constraint Optimization (KKT Framework)

1. Problem Setup

- **Constraint:** $h(x) \geq 0$

2. Feasibility Check

Determine whether the unconstrained optimum of $f(x)$ satisfies $h(x) \geq 0$.

- **If YES → Inactive Constraint**
  - **Condition:** $h(x^*) > 0$
  - **Multiplier:** $\mu = 0$
  - **Action:** Ignore the constraint boundary (the fence does not bind).
- **If NO → Active Constraint**
  - **Condition:** $h(x^*) = 0$
  - **Multiplier:** $\mu \geq 0$
  - **Action:** Treat as an equality constraint along the boundary.

> **Note for Review** — The trouble: we never know in advance which constraints are active, because that would require already knowing where the global optimum is, and if we knew that we would not need to solve anything. The next two sections build a rule that works **without** knowing the answer beforehand.

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10. Complementary Slackness

Back to the gradient equation

$$\nabla f - \sum_{i=1}^{k} \lambda_i \nabla g_i - \sum_{j=1}^{m} \mu_j \nabla h_j = 0$$

It includes a term for **every** inequality constraint. But inactive constraints must not take part in this equation, because they do not influence the optimum.

Small example. Let $k = 1$ and $m = 2$, where constraint h_2 is active and constraint h_1 is inactive. The correct equation is

$$\nabla f - \lambda_1 \nabla g_1 - \mu_2 \nabla h_2 = 0$$

and the term $\mu_1 \nabla h_1$ must be **absent**. The same effect is obtained by keeping the term but forcing its multiplier to zero: $\mu_1 = 0$.

So for each inequality constraint:

$$\mu_j = \begin{cases} 0 & \text{for inactive constraints } (h_j > 0) \\ \geq 0 & \text{for active constraints } (h_j = 0) \end{cases}$$

(for an active constraint the multiplier is normally strictly positive; the degenerate case $\mu_j = 0$ with $h_j = 0$ is also allowed, which is why the final conditions say $\mu_j \geq 0$).

One equation covers both cases. Because we cannot tell beforehand which case holds, we use the fact that in *every* case one of the two factors vanishes:

- inactive: $\mu_j = 0$,
- active: $h_j = 0$.

Hence the product is always zero:

$$\mu_j \, h_j(x^*) = 0, \qquad j = 1, \ldots, m$$

This is the **complementary slackness** condition. With it, all terms can stay in the gradient equation, and the inactive ones switch themselves off.

Situation	$h_j(x^*)$	μ_j	Product $\mu_j h_j$
Inactive	> 0	0	0
Active	0	≥ 0	0

---

11. The Sign of the Multiplier

There is one more piece of information hidden in an *active* constraint: the **direction** in which the constraint pushes.

For an active constraint the optimum lies on the boundary, and as in the equality case the gradients are parallel:

$$\nabla f = \mu \, \nabla h$$

The magnitude can differ (so we keep a scalar μ), but now the **direction** also matters:

$$\boxed{\mu \geq 0} \qquad \text{for constraints of the form } h(x) \geq 0$$

Why? The gradient ∇h points toward the region where h is larger, which is the feasible side. At a boundary optimum, the function f must *increase* as we move into the feasible region (otherwise we would walk inside and find lower values). So ∇f points into the feasible region too, in the same direction as ∇h, hence $\mu \geq 0$. If ∇f pointed the opposite way ($\mu < 0$), then $-\nabla f$ would point into the feasible region, and we could step inside and decrease f while still satisfying the constraint, which contradicts optimality.

Sketch: a family of elliptical contours of f with its center outside the feasible region, and ∇f pointing from the boundary into the feasible region.

> **Note for Review —** The sign depends on how the constraint is written. For $h(x) \geq 0$ we need $\mu \geq 0$. If a constraint is written as $h(x) \leq 0$, then ∇f must point *opposite* to ∇h at the optimum, and so $\mu \leq 0$. The easiest habit: always rewrite inequalities into the form $h(x) \geq 0$ before writing the Lagrangian. Multipliers λ_i of **equality** constraints have no sign restriction.

12. The KKT Conditions

Putting all the pieces together gives the **Karush–Kuhn–Tucker (KKT) conditions**.

Theorem 10.1 (KKT conditions). The vector $x^* \in \mathrm{R}^n$ is a critical point for minimizing f subject to $g(x) = 0$ and $h(x) \geq 0$ when there exist λ and μ such that:

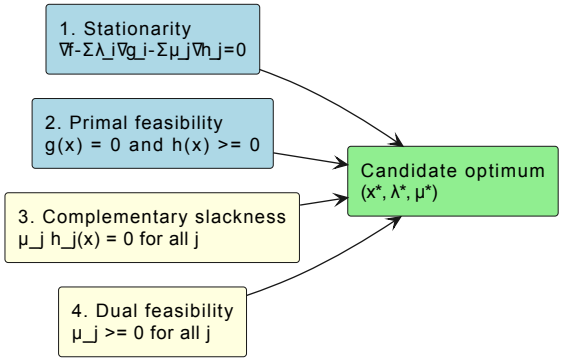
1. **Stationarity:**

$$0 = \nabla f(x^*) - \sum_i \lambda_i \nabla g_i(x^*) - \sum_j \mu_j \nabla h_j(x^*)$$

2. **Primal feasibility:** $g(x^*) = 0$ and $h(x^*) \geq 0$
3. **Complementary slackness:** $\mu_j h_j(x^*) = 0$ for all j
4. **Dual feasibility:** $\mu_j \geq 0$ for all j

(Source: Solomon, J., *Numerical Algorithms: Methods for Computer Vision, Machine Learning, and Graphics*, CRC Press, 2015. In the textbook the multiplier vectors are indexed by the sizes of g and h; here those sizes are k and m.)

Condition	What it guarantees	Where it came from
Stationarity	No feasible descent direction: gradients balance	Parallel gradients ($\nabla f = \lambda \nabla g$)
Primal feasibility	The point obeys every constraint	The problem statement itself
Complementary slackness	Inactive constraints drop out of the equation	$\mu_j = 0$ if inactive, $h_j = 0$ if active
Dual feasibility	The fence pushes back the right way	∇f points into the feasible region



To find an optimal point of a constrained problem with equalities and inequalities, we solve these conditions **together** for x^*, λ and μ.

13. KKT in Action

Let us re-run the two 1D examples through the KKT machinery, trying both possibilities for complementary slackness.

Case 2: minimize x^2 subject to $x \geq 2$

Here $h(x) = x - 2 \geq 0$ and

$$L(x, \mu) = x^2 - \mu\,(x - 2)$$

- Stationarity: $2x - \mu = 0$
- Complementary slackness: $\mu\,(x - 2) = 0$
- Dual feasibility: $\mu \geq 0$; primal feasibility: $x \geq 2$

Branch	Consequence	Verdict
$\mu = 0$ (inactive)	$2x = 0 \Rightarrow x = 0$, but $x \geq 2$ fails	Rejected (infeasible)
$x = 2$ (active)	$\mu = 2x = 4 \geq 0$, feasible	**Accepted**

So $x^* = 2$, $\mu^* = 4$. Check with the parallel-gradient picture: $\nabla f(2) = 4 = \mu\,\nabla h = 4 \cdot 1$. This also explains the earlier puzzle: the gradient of f is nonzero at the optimum because the constraint is pushing back with force $\mu = 4$.

Case 3: minimize x^2 subject to $x \geq -2$

Here $h(x) = x + 2 \geq 0$ and

$$L(x, \mu) = x^2 - \mu\,(x + 2)$$

- Stationarity: $2x - \mu = 0$
- Complementary slackness: $\mu\,(x + 2) = 0$

Branch	Consequence	Verdict
$\mu = 0$ (inactive)	$x = 0$, and $h(0) = 2 > 0$ is feasible	**Accepted**
$x = -2$ (active)	$\mu = 2x = -4 < 0$, violates dual feasibility	Rejected

So $x^* = 0$, $\mu^* = 0$: the constraint is inactive, exactly as the picture suggested.

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14. Looking Ahead to SVM

Now apply the KKT conditions to the hard-margin SVM. The variables are (θ, b), there are no equality constraints, and there are n inequality constraints

$$h_i(\theta, b) = y_i(x_i^\top \theta + b) - 1 \geq 0, \qquad i = 1, \ldots, n$$

The Lagrangian is

$$L(\theta, b, \mu) = \theta^\top \theta - \sum_{i=1}^{n} \mu_i \left[y_i(x_i^\top \theta + b) - 1\right]$$

The KKT conditions read:

- **Stationarity in θ:** $\nabla_\theta L = 2\theta - \sum_i \mu_i y_i x_i = 0 \;\Rightarrow\; \theta = \dfrac{1}{2}\sum_i \mu_i y_i x_i$
- **Stationarity in b:** $\dfrac{\partial L}{\partial b} = -\sum_i \mu_i y_i = 0$
- **Primal feasibility:** $y_i(x_i^\top \theta + b) \geq 1$
- **Complementary slackness:** $\mu_i \left[y_i(x_i^\top \theta + b) - 1\right] = 0$
- **Dual feasibility:** $\mu_i \geq 0$

Complementary slackness explains the name *support vector*. A data point can have $\mu_i > 0$ **only if** its constraint is active, i.e. $y_i(x_i^\top \theta + b) = 1$: the point lies exactly on a margin hyperplane. Every other point is inactive ($\mu_i = 0$) and does not contribute to θ at all. The solution is built only from the points sitting on the margin, which are the circled points in the first figure. The details and further simplification of these equations continue in the next lecture.

15. Quick Summary

Problem type	Condition for a candidate optimum
Unconstrained	$\nabla f(x^*) = 0$
Equality constrained, $g(x) = 0$	$\nabla f = \lambda \nabla g$ and $g(x) = 0$ (from $L = f - \lambda g$)
Inequality constrained, $h(x) \geq 0$	All four KKT conditions

Symbol	Meaning
λ_i	Multiplier of equality constraint g_i (any sign)
μ_j	Multiplier of inequality constraint $h_j \geq 0$ (must be ≥ 0)
Active	$h_j(x^*) = 0$, behaves like an equality
Inactive	$h_j(x^*) > 0$, can be ignored, $\mu_j = 0$

Storyline: SVM → quadratic program with linear inequality constraints → gradient $= 0$ is not enough → Lagrange multipliers (equalities) → active/inactive constraints → complementary slackness and sign condition → KKT.

16. Practice Questions

1. The bakery tray. A baker has exactly 20 dm of edging strip and wants to wrap it around a rectangular tray with sides x_1, x_2 so that the tray has the **largest possible area**. Rewrite this as a *minimization* problem with an equality constraint, build the Lagrangian, and solve it. At the solution, check numerically that ∇f and ∇g are parallel, and say what the sign of λ tells you about how the constraint "pushes" on the objective. Why can the Lagrange equations alone not tell you that the answer is a maximum of the area?

2. Fencing the circle. In Section 7 the equality constraint forced the point to stay *on* the unit circle. Now replace it by the inequality $x_1^2 + x_2^2 \leq 1$ and still minimize $x_1 + x_2$. Rewrite the constraint in the form $h(x) \geq 0$, list the four KKT conditions, and decide which of the two critical points from Section 7 survives and what the sign and value of μ is. Then change the objective to *maximize* $x_1 + x_2$ (write it as a minimization) and explain how the active/inactive picture and the sign of μ change. Finally, what would happen to μ if the disk were enlarged to $x_1^2 + x_2^2 \leq 100$?

3. Two fences in a field. Minimize $(x - 3)^2$ subject to $x \geq 4$ **and** $x \leq 5$. Convert both constraints into the form $h_j(x) \geq 0$, guess which one is active, and confirm your guess with all four KKT conditions (including the multipliers). Now replace the lower fence by $x \geq 1$: which multiplier becomes zero, and why does the answer agree with simply ignoring that constraint?

4. A three-point SVM, by hand. Take the one-dimensional hard-margin problem with data $x_1 = 0$ ($y_1 = -1$), $x_2 = 2$ ($y_2 = +1$), $x_3 = 3$ ($y_3 = +1$). Build $\omega = (\theta, b)$, the matrices B and A, and write out the three inequality constraints explicitly. Before solving anything, use complementary slackness and the geometry of the margin to argue which of the three multipliers *must* be zero, and then find θ^* and b^*. Which points are the support vectors, and what happens to the solution if you delete the point that is **not** a support vector?

5. The price of slack. In the soft-margin QP, besides $A\omega \geq \mathbf{1}$ there are also the constraints $\eta_i \geq 0$. Stack them under A and state the total number of rows for n data points. Consider a point that is **misclassified** with $\eta_i = 1.5$ and a point that is correctly classified **outside** the margin. For each, say which of its two constraints (margin constraint, non-negativity of η_i) is active, and what complementary slackness then forces for the corresponding multipliers. Finally, argue qualitatively what happens to the number of nonzero η_i as $c \to 0$ and as $c \to \infty$, using the structure of the cost $\theta^\top \theta + c \sum_i \eta_i$.